```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv('/content/breast-cancer.csv')
df.head()
```

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_m
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1

5 rows × 32 columns

```
df.shape
```

```
(569, 32)
```

```
df.drop(columns=['id'],inplace=True)
df.head()
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	c point
0	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0
1	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0
2	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0
3	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0
4	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0

5 rows × 31 columns

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df.iloc[:,1:], df.iloc[:,0],test_size=0.2, random_state=2)
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train,y_train)
```

```
KNeighborsClassifier
KNeighborsClassifier(n_neighbors=3)
```

```
from sklearn.metrics import accuracy_score

y_pred = knn.predict(X_test)

accuracy_score(y_test, y_pred)
```

```
0.9912280701754386
```

```
scores = []

for i in range(1,16):
```

```
    knn = KNeighborsClassifier(n_neighbors=i)

    knn.fit(X_train,y_train)

    y_pred = knn.predict(X_test)

    scores.append(accuracy_score(y_test, y_pred))

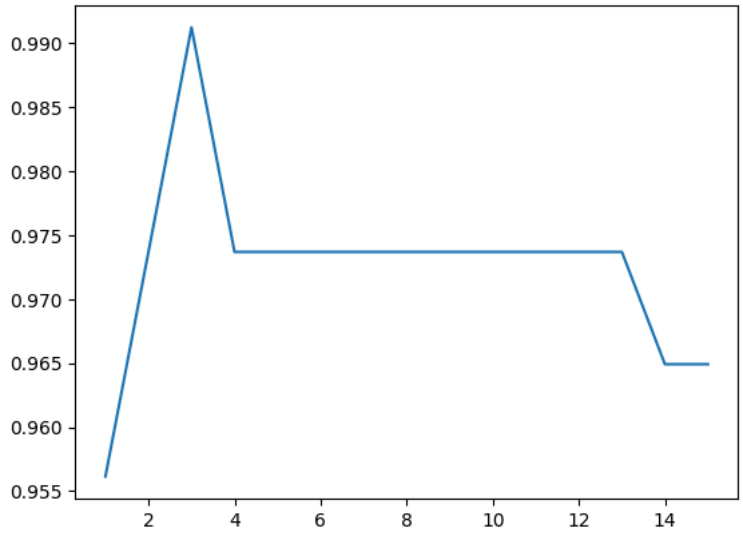
    print(accuracy_score(y_test, y_pred))
```

```
0.956140350877193
0.9736842105263158
0.9912280701754386
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9736842105263158
0.9649122807017544
0.9649122807017544
```

```
import matplotlib.pyplot as plt

plt.plot(range(1,16),scores)
```

```
[<matplotlib.lines.Line2D at 0x79453760e210>]
```




```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from joblib import dump

# Create final KNN model
knn_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('knn', KNeighborsClassifier(n_neighbors=3))
])

# Train the complete pipeline
knn_pipeline.fit(X_train, y_train)

# Save model
dump(knn_pipeline, "KNNModel.joblib")

print("KNN model saved successfully!")
```

```
KNN model saved successfully!
```

```
Start coding or generate with AI.
```